

THE CANONICAL METRIC ON HOLOMORPHIC PAIRS OVER COMPACT NON-KÄHLER MANIFOLDS

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ABSTRACT. In this paper, we prove the solvability of the vortex equation on a holomorphic vector bundle over a compact Hermitian manifold using the continuity method, and show the Kobayashi-Hitchin correspondence for holomorphic pairs. This work extends Bradlow's Kobayashi-Hitchin correspondence over compact Kähler manifolds to compact non-Kähler manifolds.

1. INTRODUCTION

For holomorphic vector bundles over compact complex manifolds, the equivalence between the stability of vector bundles and the existence of canonical metrics is known as the Kobayashi-Hitchin correspondence. The canonical metric here is a solution of the curvature equation, specifically the Hermitian-Einstein metric:

$$i\Lambda_g F_h - \lambda \text{id}_E = 0, \quad \lambda \in \mathbb{R}.$$

The proof of the Kobayashi-Hitchin correspondence developed through several key contributions. Kobayashi [8, 9, 10] and Lübke [12] first showed that the stability of vector bundles follows from the existence of a canonical metric. Later, Donaldson [3, 4, 5] and Uhlenbeck-Yau [18] proved that a stable vector bundle admits a canonical metric by taking different differential geometric approaches. Donaldson used the heat flow method, while Uhlenbeck-Yau used the continuity method.

After the establishment of this Kobayashi-Hitchin correspondence, the following extensions have been studied:

- (1) Extension to compact Gauduchon manifolds (which are, by definition, n -dimensional compact complex manifolds with a Hermitian metric ω_g satisfying $\partial\bar{\partial}\omega_g^{n-1} = 0$).
- (2) Extension to vector bundles with additional structures (Higgs fields or holomorphic sections) over compact Kähler manifolds.

Li-Yau [11] demonstrated the first extension using the continuity method. For the second extension, the case with Higgs fields was developed by Simpson [17], and the case with holomorphic sections was investigated by Bradlow [2]. In particular, Bradlow defined the notion of the canonical metric as a solution of the vortex equation and the stability for holomorphic pairs (E, ϕ) (consisting of a holomorphic vector bundle and a holomorphic section) over compact Kähler manifolds. He then proved the Kobayashi-Hitchin correspondence for holomorphic pairs using the heat flow method.